

English
Medium



EDU TERIA

72nd BPSC

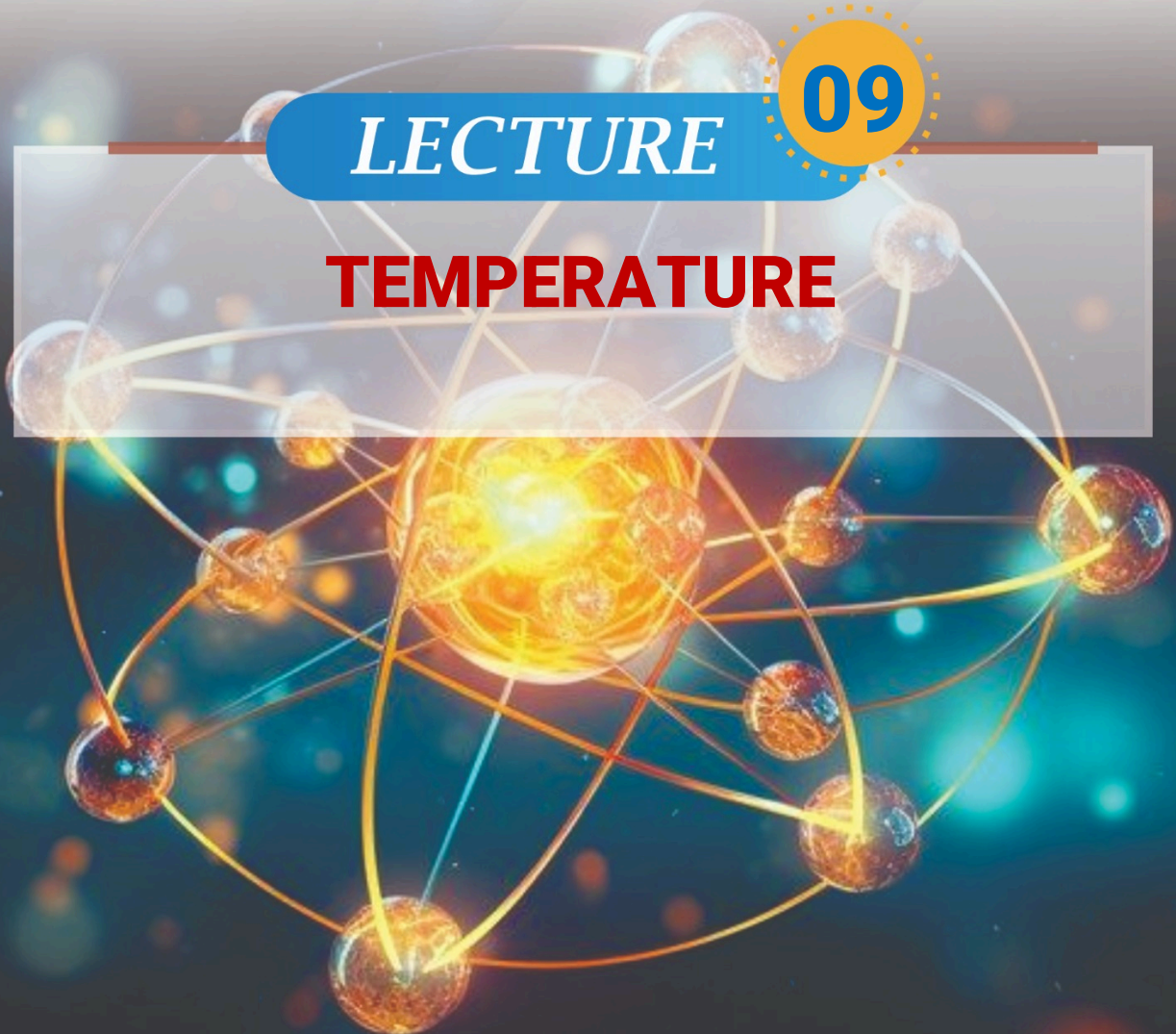
Prelims Online Batch

PHYSICS

LECTURE

09

TEMPERATURE



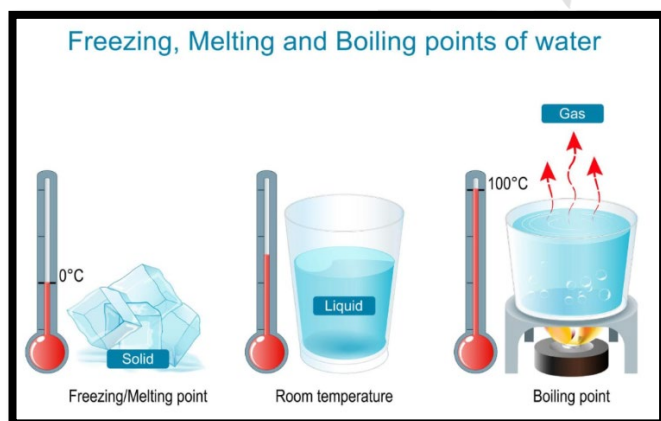
TEMPERATURE

Temperature

Conduction	Convection	Radiation
❑ Depends upon temperature difference	❑ Depends upon density difference	❑ Heat transfer without any medium
❑ Electrons transfer their vibration from one end to another without their actual movement.	❑ Transfer of heat occurs due to actual motion of particle.	❑ Due to electro magnetic radiation.
❑ Slow process	❑ Slow process	❑ Fast process at speed of 3×10^8 m/s
❑ Irregular path	❑ Irregular path	❑ Straight line pathway just like light.

Melting → Conversion of solid into its liquid form at constant temperature. (**Endothermic**).

Freezing → Reverse of Melting process with the release of energy known as Freezing. (**Endothermic process**.)



Evaporation → Conversion of surface liquid into its vapor form at all temperatures known as evaporation. Higher the temperature difference, higher will be the evaporation.

Boiling → Evaporation within the whole mass at constant temperature known as boiling. The temperature at which vapour pressure of liquid

equals to atmospheric pressure then boiling start in matter.

Sublimation– Conversion of solid directly into its vapor form **E.g**- Solid CO_2 , Camphor

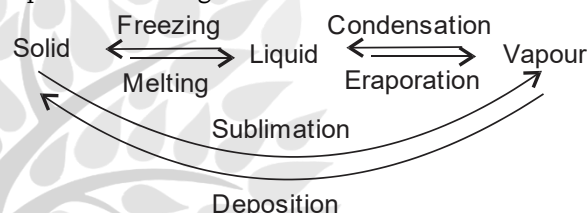
Deposition– Reverse of sublimation known as deposition.

Condensation– Conversion of vapour into its liquid from known as condensation.

Note: When impurities are added in liquid results in rise of its boiling point.

Change of State of matter:

Phase conversion due to hidden heat/ latent heat represents change in state of matter.



Temperature

- Temperature is the degree of hotness or coldness of a body.
- It determines the direction of heat flow.
- Heat always flows from higher temperature to lower temperature.

SI Unit

- Kelvin (K) — SI unit
- Other units:
 - ❖ Degree Celsius ($^{\circ}\text{C}$)
 - ❖ Degree Fahrenheit ($^{\circ}\text{F}$)

Temperature Scale	Low Point (Freezing Point)	High Point (Boiling Point)
1. Celsius	0°	100°
2. Kelvin	273.15k	373.15k
3. Fahrenheit	32°	212°
4. Reamer	0°	80°
5. Rankine	491.67°	671.67°

Measuring Instrument

Accurate measurement of temperature is essential in fields such as science, medicine, industry, and daily life. To measure temperature, various instruments—collectively known as thermometers—are used,

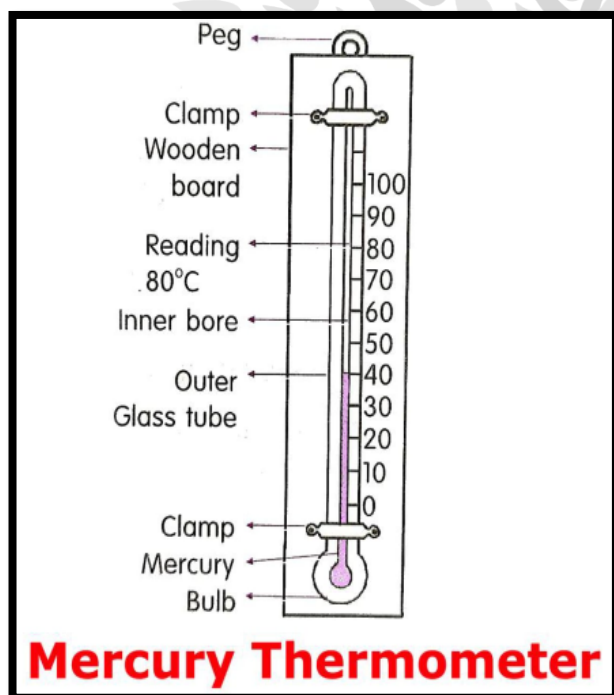


which operate on different physical principles such as thermal expansion, electrical resistance, and radiation. The development of temperature measuring instruments has enabled precise observation and control of thermal conditions, making them indispensable in both practical applications and scientific research

Types of Thermometer

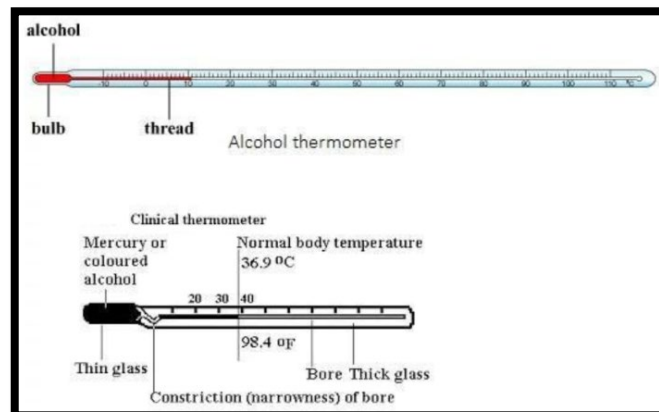
☐ Mercury Thermometer:

- **Principle:** Uses mercury's uniform expansion in a sealed glass tube.
- **Range:** Wide, from -38°C to 356°C (can be extended).
- **Pros:** Opaque silver doesn't wet glass (clear readings), good conductivity, accurate.
- **Cons:** Toxic, hazardous if broken; less used clinically now.



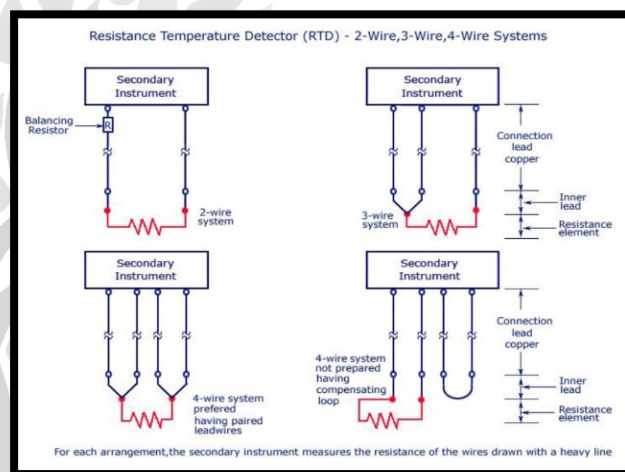
☐ Alcohol Thermometer:

- **Principle:** Colored alcohol (ethyl alcohol, etc.) expands in a glass tube.
- **Range:** Very low temps (down to -200°C), limited by alcohol's boiling point (around 80°C).
- **Pros:** Non-toxic, safe, great for very cold conditions (freezers).
- **Cons:** Transparent (needs dye), can evaporate, less durable, wets glass.



Resistance Thermometer (RTD):

- **Principle:** Measures change in electrical resistance of a conductor (like platinum) with temperature.



- **Use:** High accuracy, wide range, good linearity, common in labs.

Principle of RTD

Resistance of metals increases with increase in temperature

Common material: Platinum (Pt100, Pt1000)

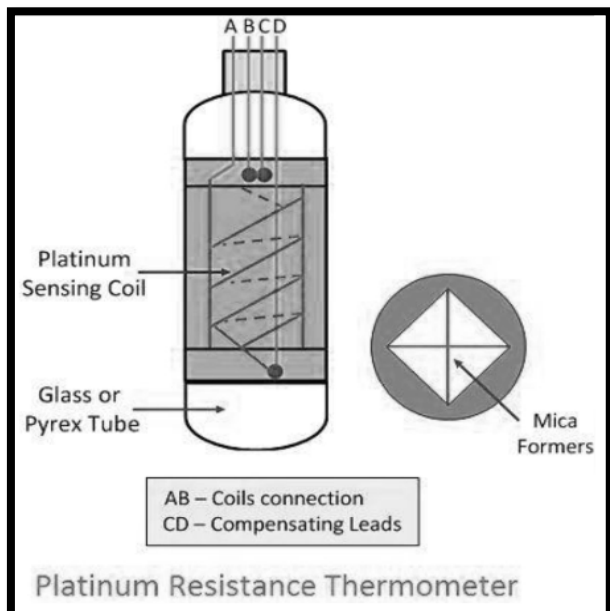
Mathematical relations:

$$R = R_0 (1 + \alpha T)$$

- R = resistance at temperature T
- R_0 = resistance at 0°C
- α = temperature coefficient

Platinum Resistance Thermometer

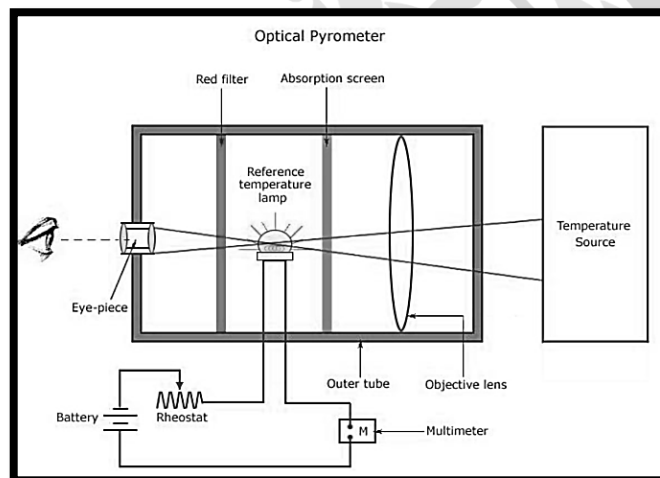
- A Platinum Resistance Thermometer (PRT) is a highly accurate temperature measuring instrument that uses platinum metal whose electrical resistance changes predictably with temperature. It is one of the most precise thermometers used in science and industry.



Pyrometer (Infrared Thermometer):

A Pyrometer is a temperature measuring instrument used to measure very high temperatures without direct contact with the object. It is especially useful where ordinary thermometers cannot be used.

Principle: Detects infrared energy (heat) emitted by objects, no contact needed.



- **Use:** Non-contact measurement, good for moving objects, high-temp industrial settings, body temp screening.

Other Types Thermometer:

- **Digital Thermometer:** Uses electronic sensors (thermistors, thermocouples), display digitally, easy for home/clinical use.
- **Bimetallic Strip Thermometer:** Two metals bonded, bend with temp changes; used in ovens, thermostats.

- **Thermocouple:** Two different metals joined; voltage changes with temp; high/low industrial temps.
- **Gas Thermometer:** Measures temp via gas pressure/volume (e.g., constant pressure gas thermometer).

Effects of Temperature

- Expansion and contraction of substances
- Change in physical state (solid $\longleftrightarrow$ liquid $\longleftrightarrow$ gas)
- Increase in molecular kinetic energy

Temperature Scales Comparison

Scale	Freezing Point of Water	Boiling Point of Water
Celsius	0°C	100°C
Fahrenheit	32°F	212°F
Kelvin	273 K	373 K

Conversion Formulae

$$\frac{C}{100} = \frac{F - 32}{180} = \frac{K - 273}{100}$$

$$\frac{C}{5} = \frac{F - 32}{9} = \frac{K - 273}{5}$$

$$C = \frac{5}{9}(F - 32) \quad K = C + 273$$

$$\frac{C - 0}{100} = \frac{F - 32}{212 - 32} = \frac{K - 273.16}{373.16 - 273.16}$$

$$T(K) = t(^{\circ}C) + 273.16$$

